

Solution Requirements

Date	15 July 2026
Team ID	xxxxxxx
Project Name	AI Debt Settlement System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users can securely register, log in, and log out using JWT authentication.	High
2	Authorization Levels	Each authenticated user can access only their own loans, dashboard, settlements, and profile information.	High
3	Loan Management	Users can add, edit, delete, search, and view loan records.	High
4	Dashboard Analytics	The system calculates Total Loans, Total Debt, EMI Ratio, Monthly Surplus, Financial Health Score, and Debt Stress.	High
5	AI Debt Settlement Recommendation	The system uses Google Gemini AI to generate personalized debt settlement recommendations based on the user's financial information.	High

6	User Profile Management	Users can view their profile information along with total loans and total outstanding debt.	Medium
7	Search & Filtering	Users can search loan records by borrower name for quick access.	Medium
8	Data Storage	Loan details, user information, and financial records are securely stored in the database.	High
9	Reporting & Insights	The system displays financial summaries and AI-generated recommendations for informed decision-making.	Medium
10	REST API Integration	React frontend communicates with FastAPI backend through RESTful APIs.	High

Step 2: Non-Functional Requirements (NFR)

S.No	Requirement Category	Requirement Description	Priority
1	Performance	Dashboard and API responses should be returned within 2 seconds under normal load.	High
2	Security	Passwords are encrypted using bcrypt, and all protected APIs require JWT authentication.	High
3	Reliability	The application should provide consistent and accurate AI recommendations without data loss.	High

4	Availability	The system should remain available during normal working hours with minimal downtime.	Medium
5	Scalability	The architecture should support adding more users, loans, and AI services in the future.	High
6	Maintainability	The project should follow a modular architecture with reusable React components and FastAPI services.	High
7	Usability	The user interface should be responsive, intuitive, and easy to navigate on desktop and mobile devices.	High
8	Compatibility	The application should work on modern web browsers such as Chrome, Edge, Firefox, and Safari.	Medium
9	Data Integrity	Each user's loan records and profile information should remain isolated and accurate.	High
10	AI Response Quality	AI-generated settlement recommendations should be relevant, understandable, and personalized based on user financial data.	High

3	Usability	The application should provide a simple, responsive, and user-friendly interface that works across desktops and laptops.	High
4	Reliability	The system should maintain accurate loan records and provide consistent AI recommendations without data loss.	High
5	Scalability	The application should support increasing numbers of users and loan records with minimal performance degradation.	Medium
6	Maintainability	The project should follow a modular architecture (React Frontend, FastAPI Backend, SQLite Database) to simplify maintenance and future enhancements.	Medium